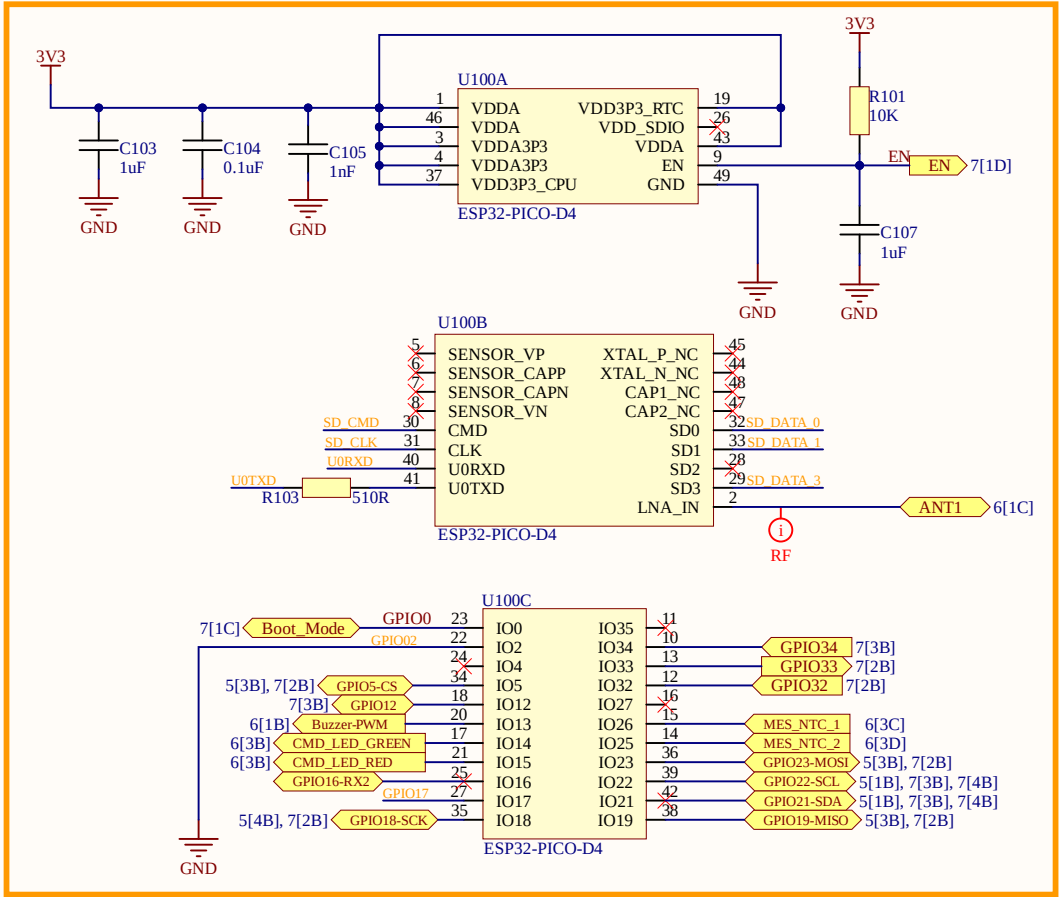
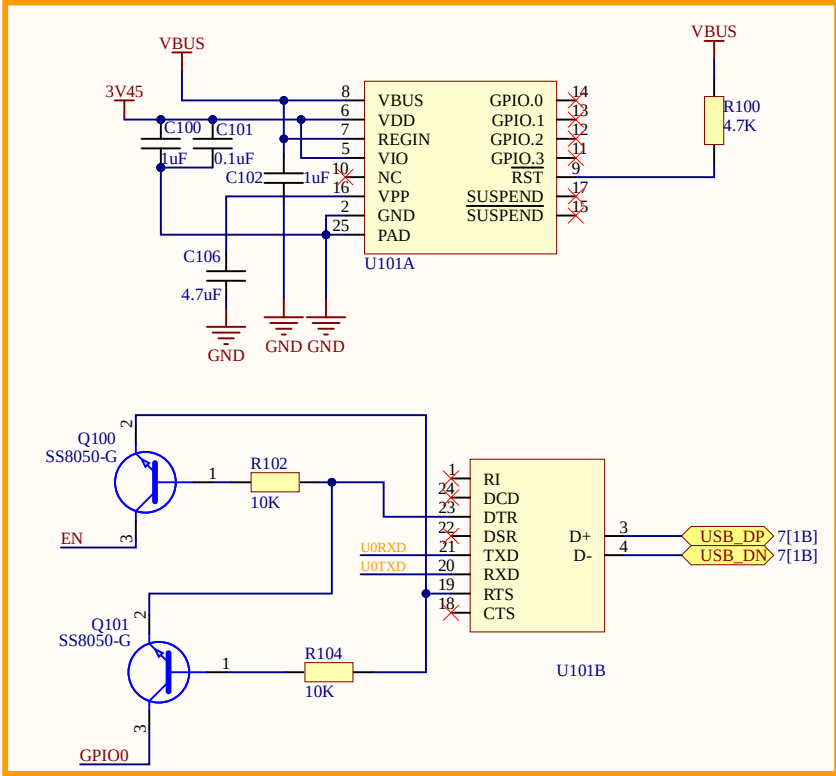


Microcontrôleur et contrôleur USB

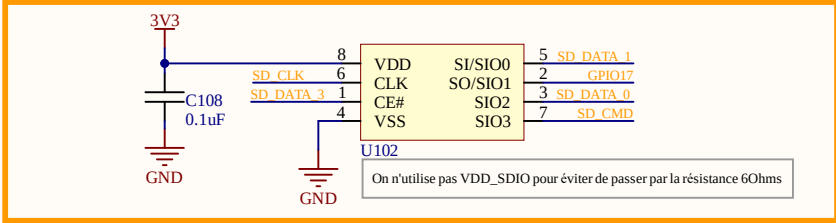
ESP32



Contrôleur USB à UART



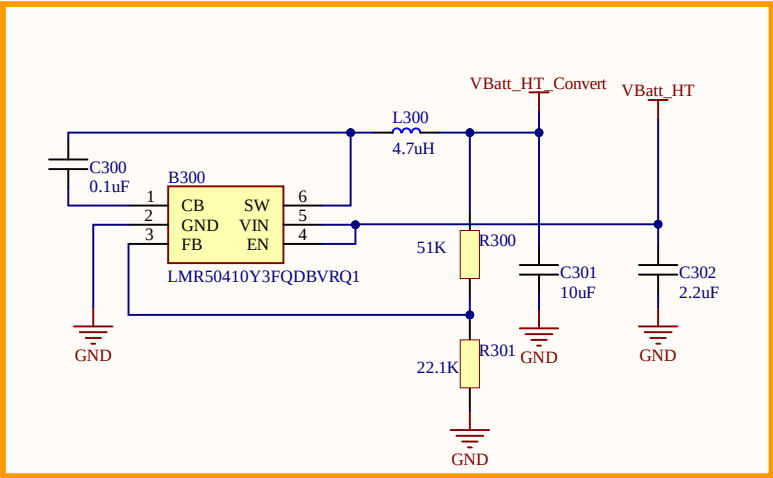
PSRAM



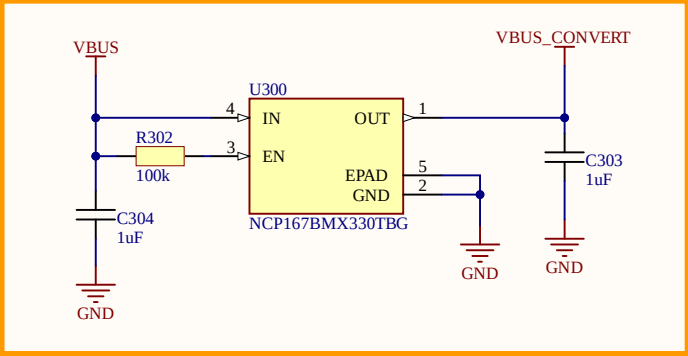
Title			
Projet M1 2024			
Size	Number		Revision
A4			V1.0 Rev A
Date:	4/10/2025	Sheet 1 of 7	
File:	S:\Ventec\microcontrôleur.SchDoc	Drawn By: J.CLEMENT & D.FOURNIER	

Alimentations USB, BT et HT

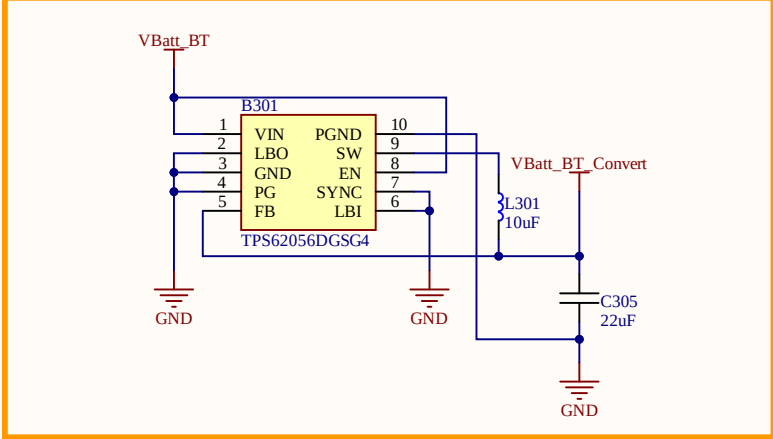
Conversion de la tension HT en 3,3 V



Conversion de la tension USB en 3,3 V



Conversion de la tension BT en 3,3 V



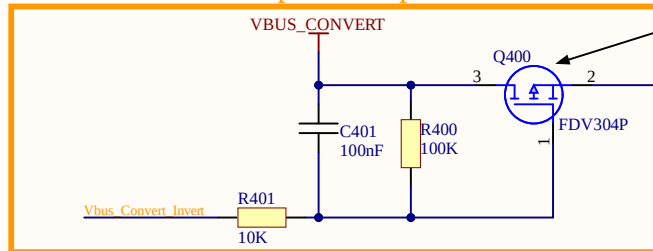
Title		
Projet M1 2024		
Size	Number	Revision
A4		V1.0 Rev A
Date:	4/10/2025	Sheet 2 of 7
File:	S:\Ventec\alimentation.SchDoc	Drawn By: J.CLEMENT & D.FOURNIER

Logique hardware : pilotage des alimentations

Table de priorité des alimentations

- 0 - Alimentation provenant de la source USB
- 1 - Alimentation provenant de la source BT
- 2 - Alimentation provenant de la source HT

Ordre de priorité 0 pour l'alimentation USB



Résistance interne 1R

Calcul de capacité de charge

Pour une consommation de 100mA sur 10ms

$$Q = I * T = 10^{-3} C$$

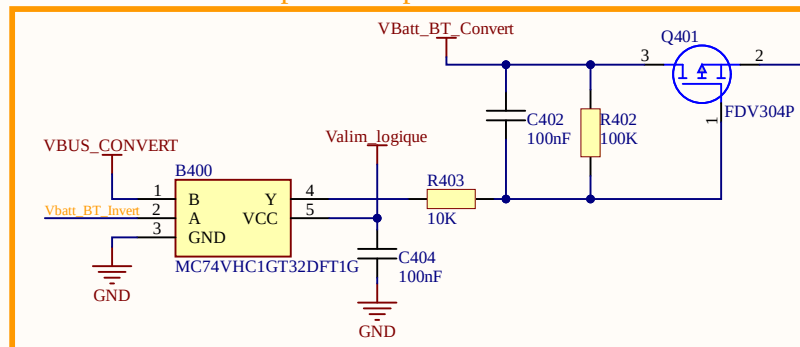
$$Q = CU = C * (\Delta U)$$

$$C = Q / (\Delta U) = (10^{-3}) / 0.2 = 5 \text{ mF}$$

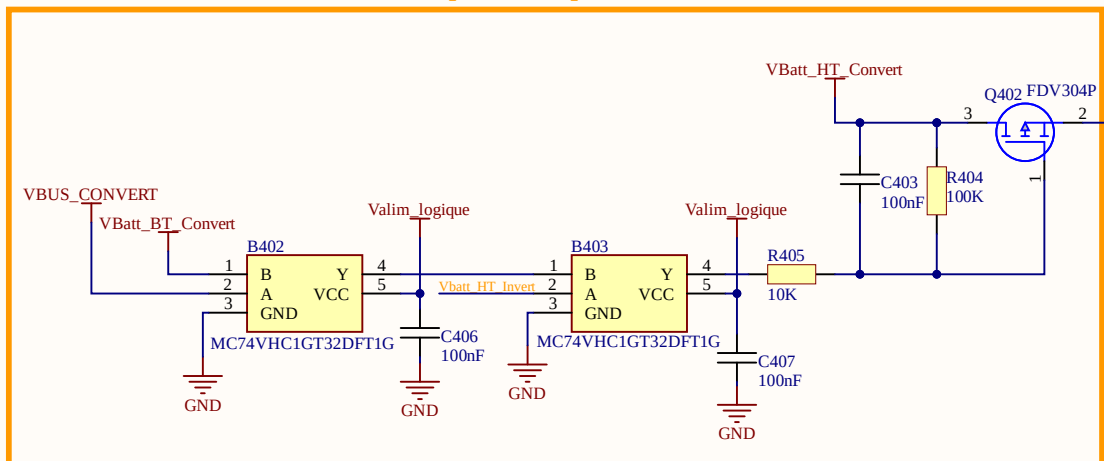
Résistance de précharge

Une résistance de précharge à ce niveau va limiter le courant à la sortie du MOS

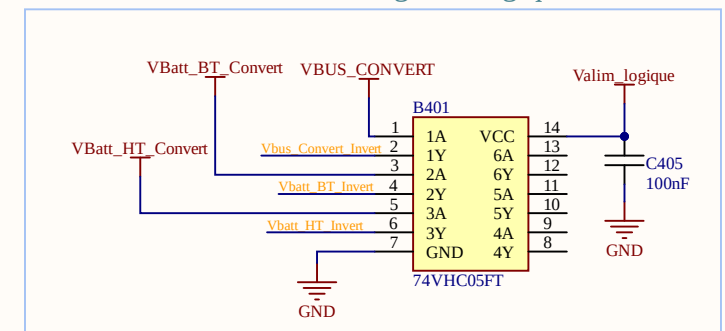
Ordre de priorité 1 pour l'alimentation BT



Ordre de priorité 2 pour l'alimentation HT

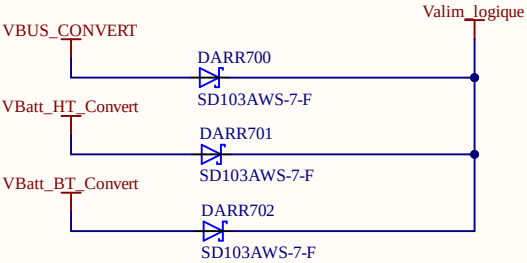


Inversion des signaux logiques



Title		
Projet M1 2024		
Size	Number	Revision
A4		V1.0 Rev A
Date:	4/10/2025	Sheet 3 of 7
File:	S:\Ventec\...logique hardware.SchDoc	Drawn By: J.CLEMENT & D.FOURNIER

Alimentation logique hardware



Title		
Projet M1 2024		
Size	Number	Revision
A4		V1.0 Rev A
Date:	4/10/2025	Sheet4 of 7
File:	S:\Ventec\..alim_logique_hard.SchDoc	Drawn By: J.CLEMENT & D.FOURNIER

4

A

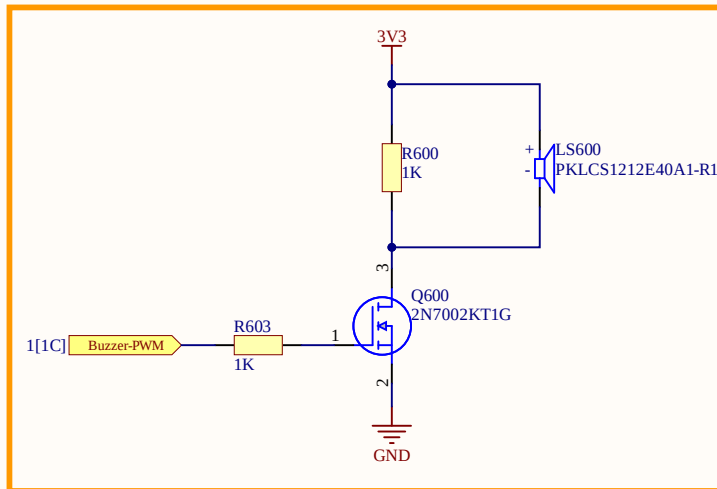


1

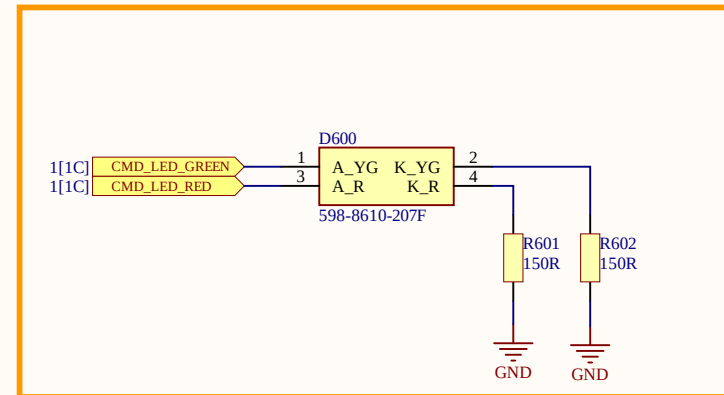
D

Interface Homme-Machine (HMI)

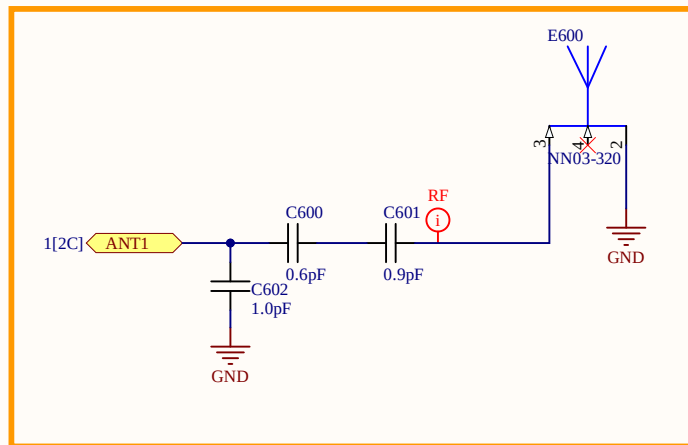
Pilotage buzzer



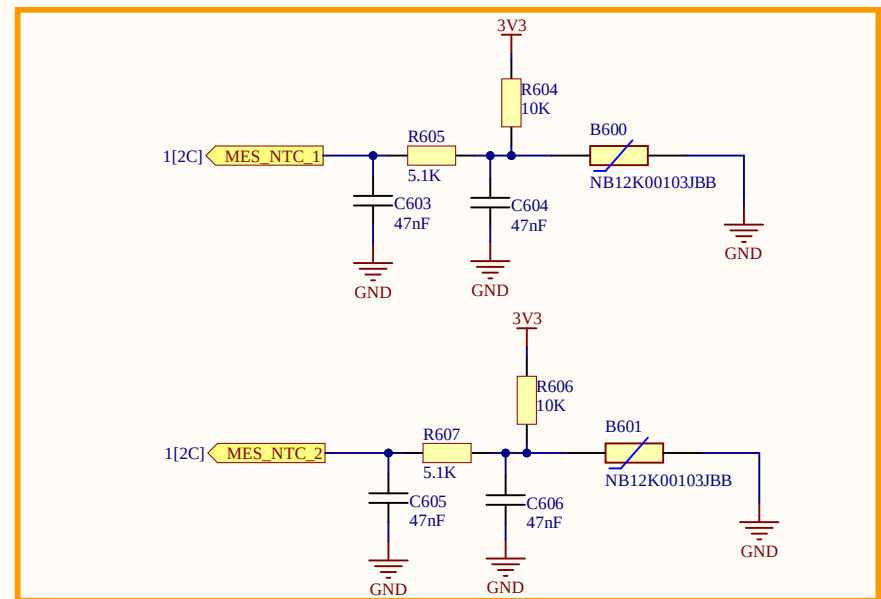
Pilotage led



Antenne pour la gestion du Bluetooth



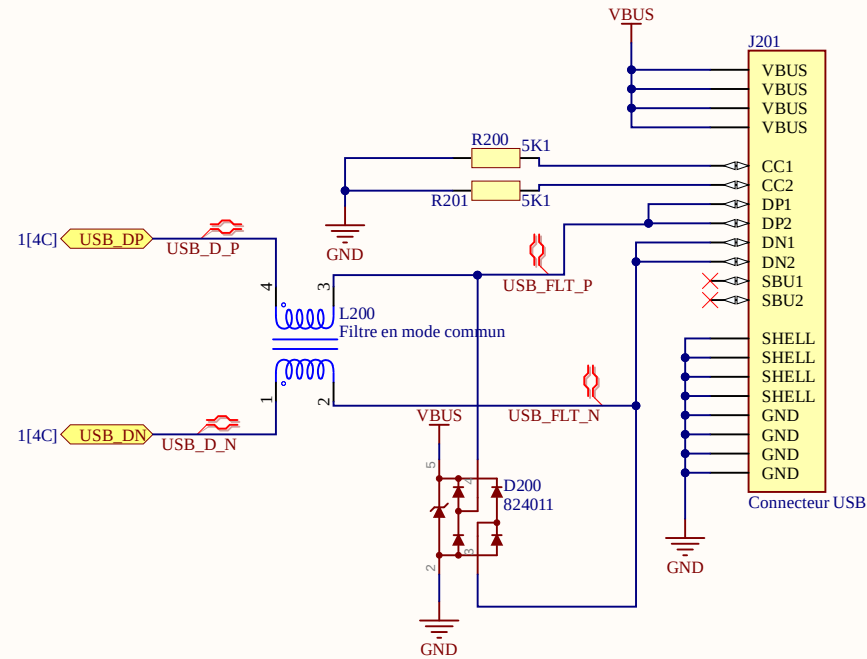
Mesures CTN



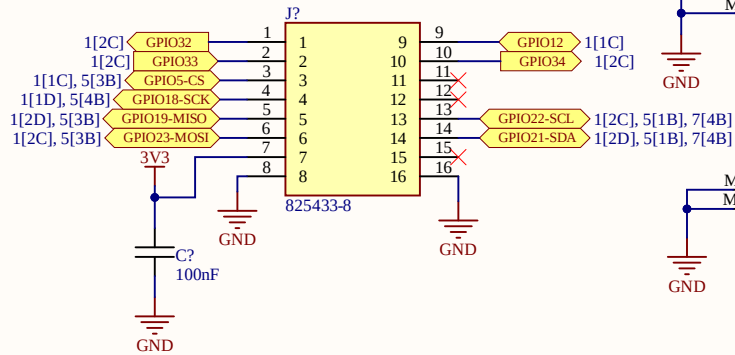
Title		
Projet M1 2024		
Size	Number	Revision
A4		V1.0 Rev A
Date:	4/10/2025	Sheet6 of 7
File:	S:\Ventec\...\hmi.SchDoc	Drawn By: J.CLEMENT & D.FOURNIER

Connecteurs

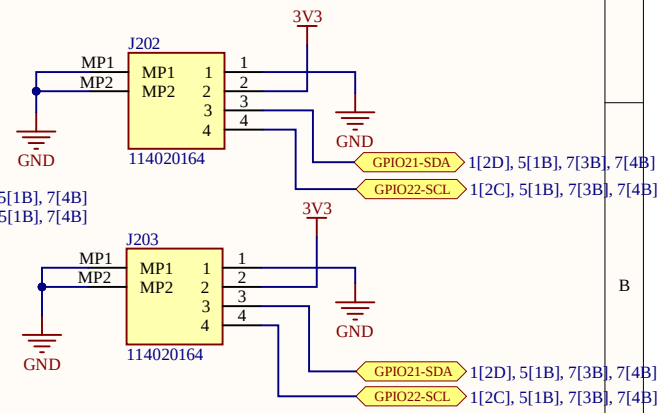
Connection USB



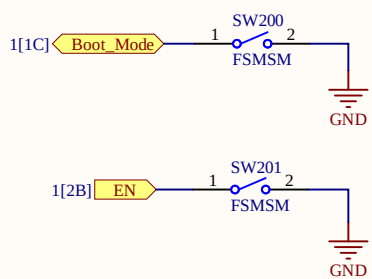
Connection Mikrobus



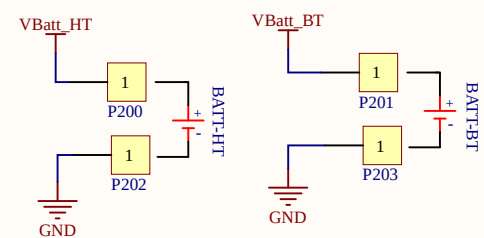
Connection Grove



Bouton reset / boot



Alimentations BT et HT



Title		
Projet M1 2024		
Size	Number	Revision
A4		V1.0 Rev A
Date:	4/10/2025	Sheet 7 of 7
File:	S:\Ventec\connecteurs.SchDoc	Drawn By: J.CLEMENT & D.FOURNIER

Board Stack Report